

MUHAMMAD AZKA HAFIZHA FAUZAN

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PROFILE SUMMARY

A Physics graduate specializing in data analysis and machine learning. As a physicist, trained to approach complex problems analytically and quantitatively — a foundation applied in building deep learning architectures, automating data pipelines, and developing computational models for large-scale datasets. Ready to contribute to the development of innovative and creative technology and digital solutions.

EDUCATION

Universitas Pendidikan Indonesia – Bandung, Indonesia Bachelor of Science in Physics GPA: 3.41 / 4.00 Focus: Computational Physics, Astrophysics, Machine Learning	2021 – 2026
SMA PGII 1 Bandung Science Program	2017 – 2020

RELEVANT EXPERIENCE

Lead Researcher – Machine Learning Project (Thesis) Universitas Pendidikan Indonesia, Bandung - Built and automated a data pipeline to process tens of thousands of SDO/NASA satellite image datasets using Python, applying a deep learning architecture (SimCLR + YOLOv5) for automated pattern extraction. - Applied self-supervised learning and object detection to generate predictions from large-scale observational data.	2025 – 2026
Lead Researcher – Simulation & Computational Modeling Universitas Pendidikan Indonesia, Bandung Developed computational models and numerical simulations to analyze complex systems, translating physical phenomena into quantitatively testable mathematical models.	2024
Lead Researcher – Environmental Data Analysis Universitas Pendidikan Indonesia, Bandung Analyzed and visualized environmental datasets using statistical approaches to generate quantitative insights as a basis for decision-making.	2024

ORGANIZATIONAL & LEADERSHIP EXPERIENCE

Team Leader – Student Creativity Program (PKM-KC) Physics Student Association, FPMIPA UPI - Led a cross-disciplinary team in developing and submitting a technology innovation proposal (Vacuum Rotary Control System based on Microcontroller) to university-level selection. - Managed technical planning, team coordination, and project documentation from concept through the selection stage.	2024
Research Team Leader – Community Service Program (PPM) Physics Student Association, FPMIPA UPI Led an invited research team in a community service program, producing a functional prototype of a coffee husk processing tool based on Vacuum Rotary & Microcontroller systems.	2023

Chair of Annual Student Assembly, HMF FPMIPA UPI

2022

Physics Student Association, UPI

- Chaired the annual student assembly and coordinated strategic decision-making processes for the organization.

Secretary

Mar 2023 – Mar 2024

Cakrawala Special Activity Unit, UPI

- Managed documentation, administrative affairs, and coordination of organizational programs.

Physics Laboratory Assistant

Even Semester 2022/2023

Dept. of Physics Education, FPMIPA UPI

- Guided students through basic physics experiments, laboratory data analysis, and lab report evaluation.

CERTIFICATION

Microsoft Office Specialist – Microsoft Word

2025 – 2030

Microsoft Corporation | Valid: 4 October 2025 – 4 October 2030

SKILLS

Programming: Python (pandas, NumPy, scikit-learn, PyTorch/TensorFlow)

Data Analysis: Time-series analysis, statistical analysis, data visualization, data pipeline

Machine Learning: Deep learning, self-supervised learning (SimCLR), object detection (YOLOv5)

Computational Modeling: Numerical simulation, complex systems modeling

Software: Microsoft Office (Word – MOS Certified), Adobe Photoshop, Adobe Premiere Pro

Soft Skills: Analytical thinking, problem solving, team leadership, adaptability